

ABHISHEK KUMAR GUPTA

Gwalior, Madhya Pradesh, India

+91-9956819529

abhishekgupta140205@gmail.com

LinkedIn

GitHub

Education

Rustamji Institute of Technology (RJIT)

B.Tech in Information Technology

2024 – 2028

Gwalior, India

Army Public School, Kunraghat

Class XII (CBSE)

2023 – 2024

Gorakhpur, India

Internship

Robotics Intern | AatonovaZ Technologies Private Limited

08 Sept 2026 – Present

- Working with the **hardware team** on the development of a **drone flight controller (FC)** and related hardware systems.

Jawahar Navodaya Vidyalaya

Robotics Trainer

05 Nov 2025 – 27 Nov 2025

Nalanda, Bihar

- Trained student teams in **Arduino**, **sensors**, and core **robotics fundamentals** through hands-on **embedded systems** sessions.
- Mentored teams for robotics competitions, guiding design and debugging of sensor-driven, competition-ready builds.

Motion Cut

Frontend Web Developer Intern

Remote

- Assisted in frontend development and UI improvements using **HTML**, **CSS**, and **JavaScript**, collaborating with the development team.

Projects

NIDAR AirMouse Project | Raspberry Pi, LiDAR, SLAM, Computer Vision, UAV

2026 – Present

- Developing an autonomous UAV navigation system for **GPS-denied environments** using a **Raspberry Pi** as the onboard compute platform.
- Integrating **LiDAR** for depth perception and a **computer vision** pipeline for object and person detection.
- Exploring **SLAM fundamentals** for environmental mapping, and testing Raspberry Pi communication with the flight-control system.

Micromouse Autonomous Robot | C++, PID, VL53L0X, Encoders, TB6612FNG

2025

- Designed a high-speed autonomous maze-solving robot using **PID control** and path-planning algorithms on **Arduino Mega**.
- Integrated **VL53L0X ToF sensors** and encoder feedback with **TB6612FNG** motor drivers for real-time maze navigation.

Autonomous Drone / UAV | Flight Stabilization, Sensors

2024

- Implemented basic **flight stabilization** concepts and **sensor integration** for steady flight, building foundational UAV architecture knowledge.

Autonomous Utility Robot & RC Plane | Embedded C, CAD

2024

- Built an embedded, autonomous line-following utility robot and designed a fixed-wing **RC aircraft** using **CAD**.

Technical Skills

Programming: C++, Python, Embedded C, Arduino, HTML, CSS, JavaScript, SQL

Embedded & Hardware: Raspberry Pi, Arduino, ESP, Sensors, LiDAR, ToF Sensors, Encoders, Motor Drivers, GPIO, UART, I2C, SPI, PCB Design

Robotics & UAV: UAV Systems, Autonomous Navigation, Sensor Integration & Fusion, SLAM Fundamentals, Computer Vision, Real-Time Control

Tools: Linux, Git/GitHub, Arduino IDE, CAD, 3D Printing

Achievements

2nd Position | Micromouse Competition, IIT BHU

2026

1st Position | Micromouse, IIT Patna Celesta

2025

7th Position | Micromouse, IIT Guwahati

2025

Idea Selected | Smart India Hackathon (SIH)

2024

Certifications

- AI/ML Certification
- Big Data Analytics
- Google Certification
- Oracle Cloud Infrastructure
- CAD & 3D Printing
- Team Robolution Member